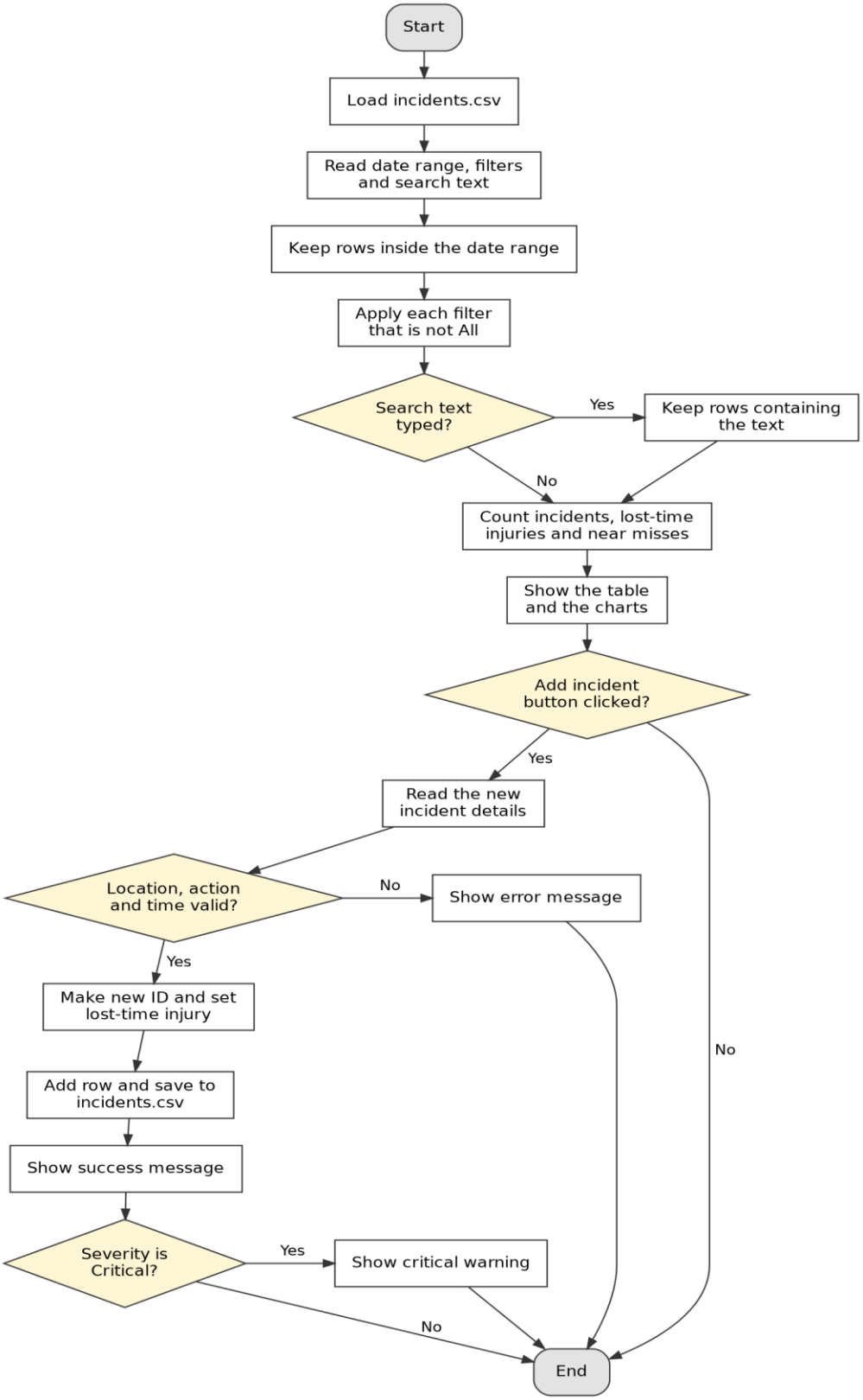
Flowchart: Part (c): Safety Incident Database



Pseudocode

```
START

    LOAD incidents FROM "incidents.csv"

    // PART 1: FILTER AND ANALYSE SAFETY INCIDENTS

    INPUT start_date, end_date, department, shift,
          incident_type, severity, search_text

    shown = rows in incidents WHERE
            date >= start_date AND date <= end_date

    IF department != "All" THEN
       shown = rows in shown WHERE department = selected department
    ENDIF

    IF shift != "All" THEN
       shown = rows in shown WHERE shift = selected shift
    ENDIF

    IF incident_type != "All" THEN
       shown = rows in shown WHERE incident_type = selected type
    ENDIF

    IF severity != "All" THEN
       shown = rows in shown WHERE severity = selected severity
    ENDIF

    IF search_text != "" THEN
       search_term = lowercase(search_text)
       filtered_rows = empty list

       FOR each row IN shown
          row_text = lowercase(all values in row joined together)

          IF search_term is contained in row_text THEN
             ADD row TO filtered_rows
          ENDIF
       ENDFOR

       shown = filtered_rows
    ENDIF

    incidents_found = number of rows in shown

    lost_time = count rows in shown
                WHERE lost_time_injury = "Yes"

    near_misses = count rows in shown
```

```
            WHERE incident_type = "Near Miss"

DISPLAY incidents_found, lost_time, near_misses
DISPLAY shown as a table

type_counts = count rows in shown grouped by incident_type
DISPLAY bar chart of type_counts

shift_counts = count rows in shown grouped by shift
DISPLAY bar chart of shift_counts


// PART 2: ADD A NEW SAFETY INCIDENT

INPUT new_date, new_time, new_shift, new_department,
      location, new_type, new_severity, injury, cause, action

IF "Add incident" button is clicked THEN

    IF location is empty OR action is empty THEN
        DISPLAY "Please fill in the location and corrective action."

    ELSE IF new_time is not a valid HH:MM time THEN
        DISPLAY "Time must be valid, e.g. 08:30."

    ELSE
        biggest = 0

        FOR each old_id IN incidents.incident_id
            number_part = numeric part of old_id

            IF number_part is valid AND number_part > biggest THEN
                biggest = number_part
            ENDIF
        ENDFOR

        new_id = "INC-" + (biggest + 1 formatted as 4 digits)

        IF injury = "Lost Time Injury" THEN
            lti = "Yes"
        ELSE
            lti = "No"
        ENDIF

        new_row = record containing:
            incident_id = new_id
            date = new_date
            time = new_time
            shift = new_shift
            department = new_department
            location = location
```

```
        incident_type = new_type
        severity = new_severity
        injury_status = injury
        lost_time_injury = lti
        cause = cause
        corrective_action = action
        status = "Open"

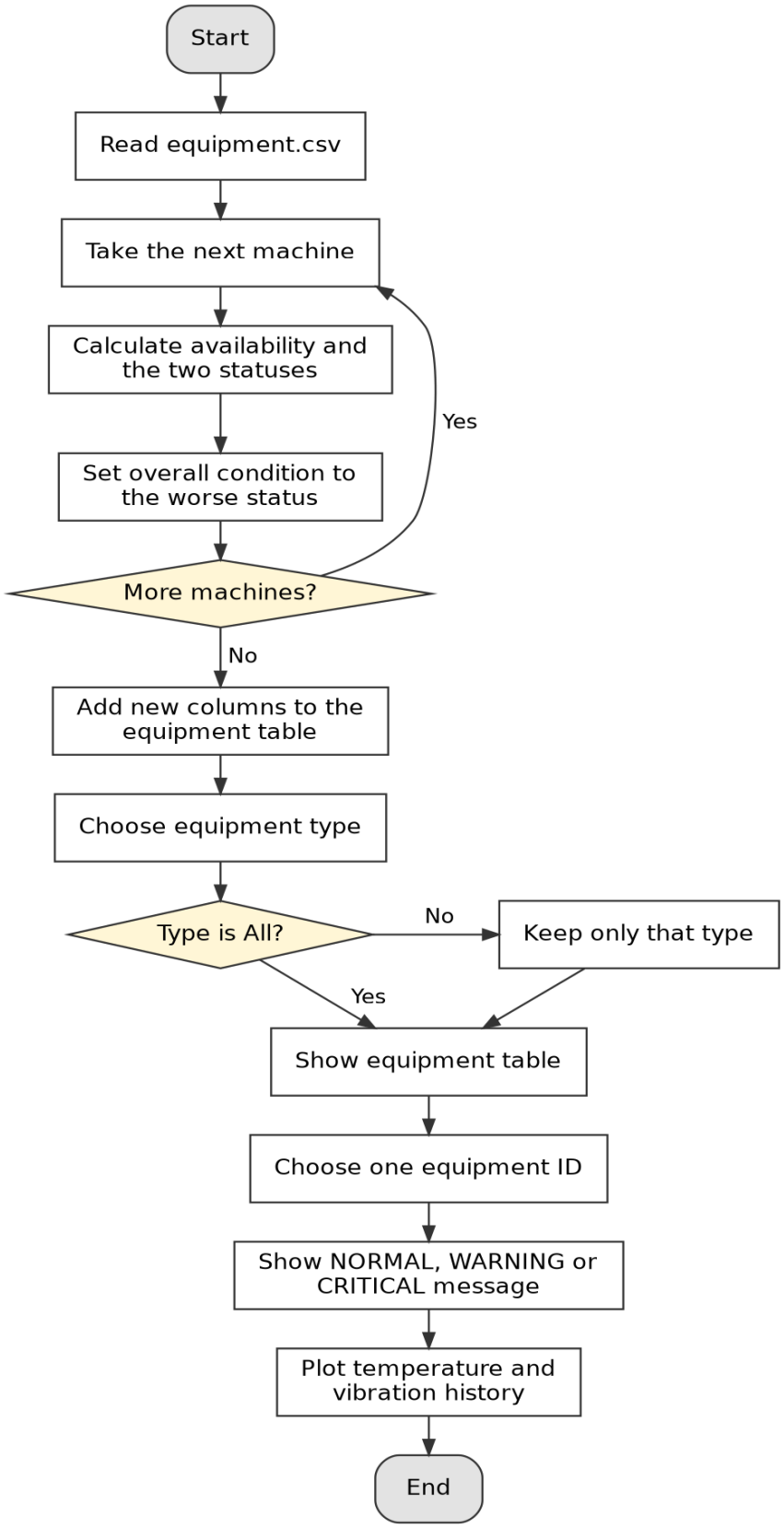
    ADD new_row TO incidents
    SAVE incidents TO "incidents.csv"

    DISPLAY "Incident saved successfully."

    IF new_severity = "Critical" THEN
        DISPLAY "CRITICAL incident. Notify management immediately."
    ENDIF


END
```

Flowchart: Part (c): Safety Incident Database



Pseudocode

```
FUNCTION get_availability(operating, downtime)
   IF operating + downtime = 0 THEN
      RETURN 0
   ELSE
      RETURN operating / (operating + downtime) * 100
   ENDIF
END FUNCTION


FUNCTION get_temp_status(temp)
   IF temp > 100 THEN
      RETURN "CRITICAL"
   ELSE IF temp >= 80 THEN
      RETURN "WARNING"
   ELSE
      RETURN "NORMAL"
   ENDIF
END FUNCTION


FUNCTION get_vib_status(vib)
   IF vib > 8 THEN
      RETURN "CRITICAL"
   ELSE IF vib >= 5 THEN
      RETURN "WARNING"
   ELSE
      RETURN "NORMAL"
   ENDIF
END FUNCTION


START

   LOAD equipment FROM "equipment.csv"

   // PART 1: CALCULATE AVAILABILITY AND EQUIPMENT CONDITION

   CREATE empty lists for availability, temp_status,
        vib_status and condition

   FOR each machine IN equipment

      operating_time = machine.operating_time
      downtime = machine.downtime
      temperature = machine.temperature
      vibration = machine.vibration

      machine_availability = get_availability(
         operating_time, downtime
```

```
    )

    machine_temp_status = get_temp_status(temperature)
    machine_vib_status = get_vib_status(vibration)

    IF machine_temp_status = "CRITICAL"
       OR machine_vib_status = "CRITICAL" THEN
        machine_condition = "CRITICAL"

    ELSE IF machine_temp_status = "WARNING"
          OR machine_vib_status = "WARNING" THEN
       machine_condition = "WARNING"

    ELSE
       machine_condition = "NORMAL"
    ENDIF

    ADD machine_availability TO availability
    ADD machine_temp_status TO temp_status
    ADD machine_vib_status TO vib_status
    ADD machine_condition TO condition

ENDFOR

ADD availability TO equipment as a new column
ADD temp_status TO equipment as a new column
ADD vib_status TO equipment as a new column
ADD condition TO equipment as a new column


// PART 2: FILTER AND DISPLAY EQUIPMENT

INPUT equipment_type

IF equipment_type = "All" THEN
    shown = all rows of equipment
ELSE
    shown = rows in equipment WHERE
          equipment_type = selected equipment_type
ENDIF

DISPLAY shown as a table

// END OF PART 2: EQUIPMENT FILTERING


// PART 3: DISPLAY EQUIPMENT STATUS AND WARNINGS

INPUT chosen_equipment_id

row = row in equipment WHERE
```

```
        equipment_id = chosen_equipment_id

    IF row exists THEN

        DISPLAY row condition

        IF row condition = "NORMAL" THEN
            DISPLAY "NORMAL: Equipment operating within defined monitoring limits."
        ENDIF

        IF row temp_status = "WARNING" THEN
            DISPLAY "WARNING: Equipment temperature approaching predefined threshold."
        ENDIF

        IF row temp_status = "CRITICAL" THEN
            DISPLAY "CRITICAL: Equipment temperature exceeds predefined threshold. Inspection
required."
        ENDIF

        IF row vib_status = "WARNING" THEN
            DISPLAY "WARNING: Equipment vibration approaching predefined threshold."
        ENDIF

        IF row vib_status = "CRITICAL" THEN
            DISPLAY "CRITICAL: Equipment vibration exceeds predefined threshold. Inspection
required."
        ENDIF


        // PART 4: DISPLAY EQUIPMENT HISTORY

        history = readings WHERE
                equipment_id = chosen_equipment_id

        DISPLAY line chart of temperature history
        DISPLAY line chart of vibration history

    ELSE
        DISPLAY "Equipment ID not found."
    ENDIF

END
```